TASK 1 - Basic
Cybersecurity Awareness & Security Audit

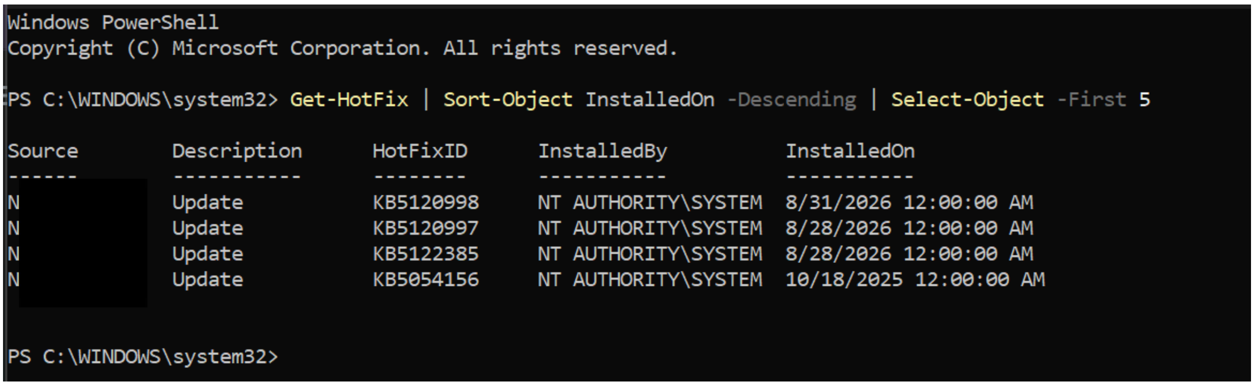
For this task I decided to use:
1. Local system
2. Brave browser
3. Windows Powershell

Step 1: Checking password practices on my own system.

I opened my browser's saved passwords list. Fortunately the passwords met the minimum criteria and were not reused anywhere else. All of the passwords contained random characters with no two same characters or characters in sequence (such as 123… or abc…).

Step 2: Checking system update status

I entered the below command to show 5 most recent security updates:
Get-HotFix | Sort-Object InstalledOn -Descending | Select-Object -First 5



If the last update is very old, this would lead to "system not updated regularly". The system needs to be updated regularly to prevent any security risks.

Step 3: Checking for exposed/open ports

The command used in the powershell for this step is:
netstat -ano | findstr LISTENING

This lists the ports on my computer that are open and listening. Any expected ports that are not recognizable are marked as a finding worth investigating.

```
PS C:\WINDOWS\system32> netstat -ano | findstr LISTENING
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       1912
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING       4
  TCP    0.0.0.0:2869           0.0.0.0:0              LISTENING       4
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING       12240
  TCP    0.0.0.0:5357           0.0.0.0:0              LISTENING       4
  TCP    0.0.0.0:7680           0.0.0.0:0              LISTENING       23272
  TCP    0.0.0.0:28414          0.0.0.0:0              LISTENING       29432
  TCP    0.0.0.0:35474          0.0.0.0:0              LISTENING       23560
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING       1592
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING       1432
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING       2448
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING       4120
  TCP    0.0.0.0:49672          0.0.0.0:0              LISTENING       5852
  TCP    0.0.0.0:49684          0.0.0.0:0              LISTENING       1552
  TCP    127.0.0.1:5354         0.0.0.0:0              LISTENING       6300
  TCP    127.0.0.1:5939         0.0.0.0:0              LISTENING       7288
  TCP    127.0.0.1:23435        0.0.0.0:0              LISTENING       2616
  TCP    127.0.0.1:27015        0.0.0.0:0              LISTENING       6268
```

From the above screenshot, some of the ports are noted below:

Port	**Services**	**Notes**
135	Windows RPC	Standard
445	SMB (file sharing)	Only a risk, if I'm using a public network
2869	SSDP/UPnP device discovery	Standard
5040	Windows Connected Devices Platform	Standard
5357	Network device API	Standard
7680	Windows update delivery optimization	Standard

28414, 35474, 49664 - 49684	Windows RPC	Standard

Step 4: Checking Firewall and User Account Status

This is done to check whether the system's firewall is actually turned on, and whether you have unnecessary admin/guest accounts sitting around.

The commands that were run are:
Get-NetFirewallProfile | Select-Object Name, Enabled
Get-LocalUser | Select-Object Name, Enabled
Get-LocalGroupMember -Group "Administrators"

```
PS C:\WINDOWS\system32> Get-NetFirewallProfile | Select-Object Name, Enabled

Name    Enabled
----    -------
Domain     True
Private    True
Public     True
```

```
PS C:\WINDOWS\system32> Get-LocalUser | Select-Object Name, Enabled

Name                  Enabled
----                  -------
Administrator           False
CodexSandboxOffline      True
CodexSandboxOnline       True
DefaultAccount          False
Guest                   False
[redacted]              False
USER                     True
WDAGUtilityAccount      False
WsiAccount              False
[redacted]              False
                         True
```

```
PS C:\WINDOWS\system32> Get-LocalGroupMember -Group "Administrators"

ObjectClass Name                          PrincipalSource
----------- ----                          ---------------
User        [redacted] OFFICIAL\Administrator Local
User                   OFFICIAL\USER          Local
```

Windows Firewall was found enabled for the Private and Public network profiles, which is the expected secure configuration.

The built-in Guest account was disabled, which is correct. However, the audit found 4 enabled local user accounts (CodexSandboxOffline, CodexSandboxOnline, USER) versus only 1 account (hidden) that appeared to be a standard named personal account, but was disabled.

Having multiple unclear/unused accounts enabled increases the attack surface, since each is a potential entry point for an attacker if compromised. It is recommended that any accounts not actively in use be disabled, and that all enabled accounts be reviewed.

Recommendations

- Enable automatic OS updates.
- Replace weak/reused passwords with a password manager and 12+ character passwords.
- Close/investigate unused open ports.

Conclusion

This audit identified issues of varying severity. Applying the recommendations above would reduce the attack surface of the tested system (local).